Design, Automation, and Analytics Through No-Code Technologies

PROPERTY BOOKING SYSTEM

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Abstract –

In this project, I designed and developed a *Real Estate Property Visit Booking System* using AppSheet and Google Sheets as a no-code solution. The primary goal was to create a centralized, admin-controlled application that simplifies the management of property listings, agents, clients, and property visit bookings. Instead of relying on manual coordination, the system introduces a structured workflow with validation rules, approval mechanisms, automation, and dashboards. Through this project, I explored practical aspects of data modelling, business logic, and workflow automation, making the application suitable for both real-world usage and academic evaluation.

Problem Statement –

Case Study 124 :Real Estate Property Visit Booking System

Design and implement a realistic, operations-focused application that requires automation, monitoring, and structured workflows, including well-defined views, data slices for critical states, action buttons that mirror real operational decisions, visual indicators for priority and risk, and automated workflows for alerts, reminders, escalations, and status updates to ensure efficient and controlled system management.

Objectives –

The main objectives of this project were:

- To centralize property, agent, and client information in one system
- To manage property visit bookings through an admin-approval workflow
- To prevent invalid or conflicting bookings using rule-based validation
- To provide real-time operational visibility using dashboards
- To practically implement AppSheet features such as slices, actions etc…

System Architecture –

I. Technology Stack

To build this system, I used the following technologies:

- **Frontend & Logic:** AppSheet (no-code application platform)
- **Database:** Google Sheets
- **Automation:** AppSheet Bots for email notifications

This combination allowed rapid development while maintaining flexibility and control.

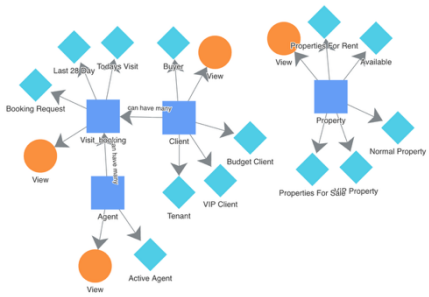
II. Data Tables

The application is built on four core tables:

- **Properties**
- **Agents**
- **Clients**
- **Visit_Bookings**

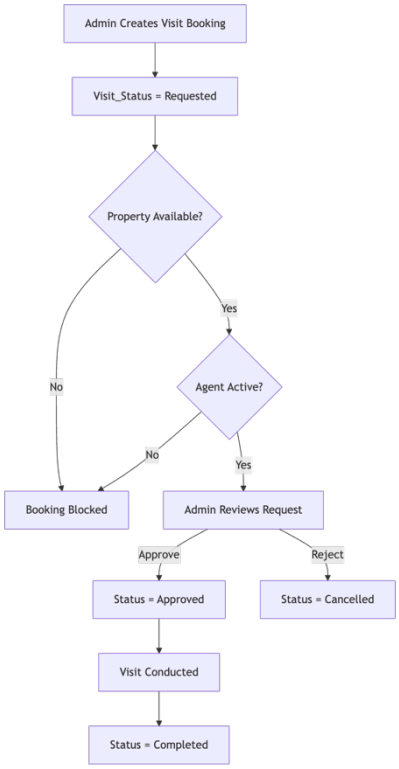
Each table uses a unique business ID as its primary key, and the tables are connected using reference relationships to maintain data integrity and enable search-by-name functionality.

Data Model & Relationships –



This structure ensures that each visit booking is always associated with exactly one property, one agent, and one client.

Application Workflow –



The workflow begins when the admin creates a visit booking. The system automatically assigns the status as *Requested*. Before approval, the application validates whether the selected property is available and whether the assigned agent is active. Based on the admin's decision, the visit is either approved or cancelled. After the visit is conducted, the status is marked as *Completed*.

Business Rules and Formulas –

I. Conditional Column Visibility

Buy Price:

```
[Transaction_Type] = "Buy"
```

Rent Price:

```
[Transaction_Type] = "Rent"
```

II. Validation Rules

Only available properties can be booked

```
SELECT(
Properties[Property_ID],
[Availability_Status] = "Available"
)
```

Only active agents can be assigne

```
SELECT(
Agents[Agent_ID],
[Active_Status] = "Yes"
)
```

Slices (Data Segmentation) –

Slice Name	Purpose	Formula
Pending_Approvals	Visits awaiting admin approval	[Visit_Status] = "Requested"
Today_Visits	Visits scheduled for today	[Visit_Date] = TODAY()
VIP_Properties	High-priority properties	[Priority] = "VIP"
Available_Properties	Properties open for booking	[Availability_Status] = "Available"

Actions (Operational Controls) –

Approve Visit:

[Visit_Status] = "Approved"

Mark Completed:

[Visit_Status] = "Completed"

Cancel Visit:

[Visit_Status] = "Cancelled"

Views and Dashboard Design –

View Name	View Type	Source Table / Slice	Purpose
Agent	Deck View	Agents	Displays active agents in a card-style layout for quick visual identification and assignment.
Client Master	Table View	Clients	Maintains a structured list of all clients with searchable and sortable details.
Book Visit	Table View	Visit_Bookings	Allows the admin to create, view, and manage property visit bookings.
Property	Table View	Properties	Serves as the master list of all properties with status, pricing, and priority details.
Dashboard	Dashboard	Multiple views	Provides a centralized operational overview of key metrics and activities.
Today's Visits	Table View	Today_Visits (Slice)	Displays all property visits scheduled for the current day.
Last 28 Days	Table View	Visit_Bookings	Shows visit activity from the past 28 days for trend and performance review.
Booking Request	Table View	Pending_Approvals (Slice)	Lists visit requests awaiting admin approval.
Active Agent	Table View	Agents (Filtered)	Shows only agents marked as active for operational assignment.

Benefits –

Through this system, I achieved:

- Centralized and structured data management
- Reduced manual coordination
- Clear approval and audit trail
- Real-time operational visibility
- A scalable foundation suitable for real-world extension

Limitations and Future Enhancements

Current Limitations –

- Single-admin operation
- Email-only notifications

Future Enhancements –

- Role-based access for agents and clients
- WhatsApp or SMS integration
- Advanced analytics and charts
- Payment or token-based visit booking

Conclusion –

Through this project, I successfully demonstrated how AppSheet can be used to design a structured, rule-driven business application without traditional coding. By combining clean data modeling, validation logic, approval workflows, automation, and dashboards, the system balances academic rigor with real-world practicality. This project strengthened my understanding of no-code application design and highlights AppSheet as a powerful platform for rapid and professional application development.

APP LINK:

https://www.appsheet.com/start/eb2ab7b5-af6e-4a44-863e-4f9a00186675

THANK YOU,